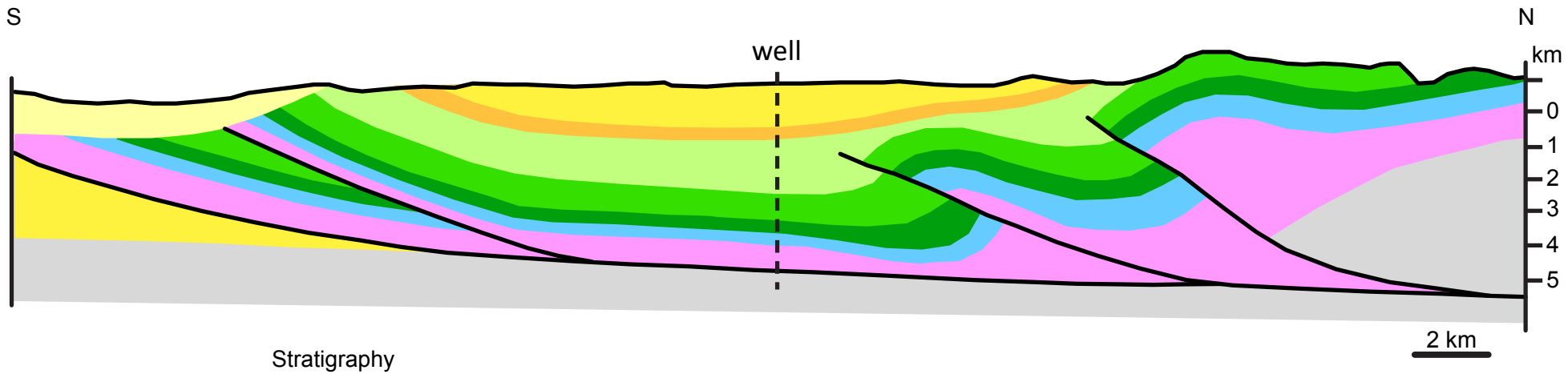
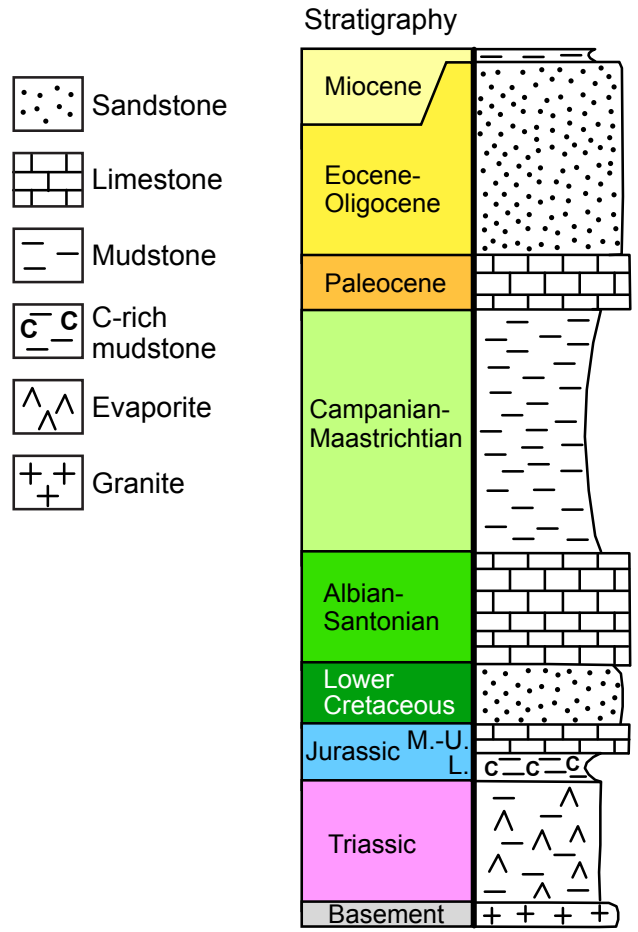
Petroleum systems exercise

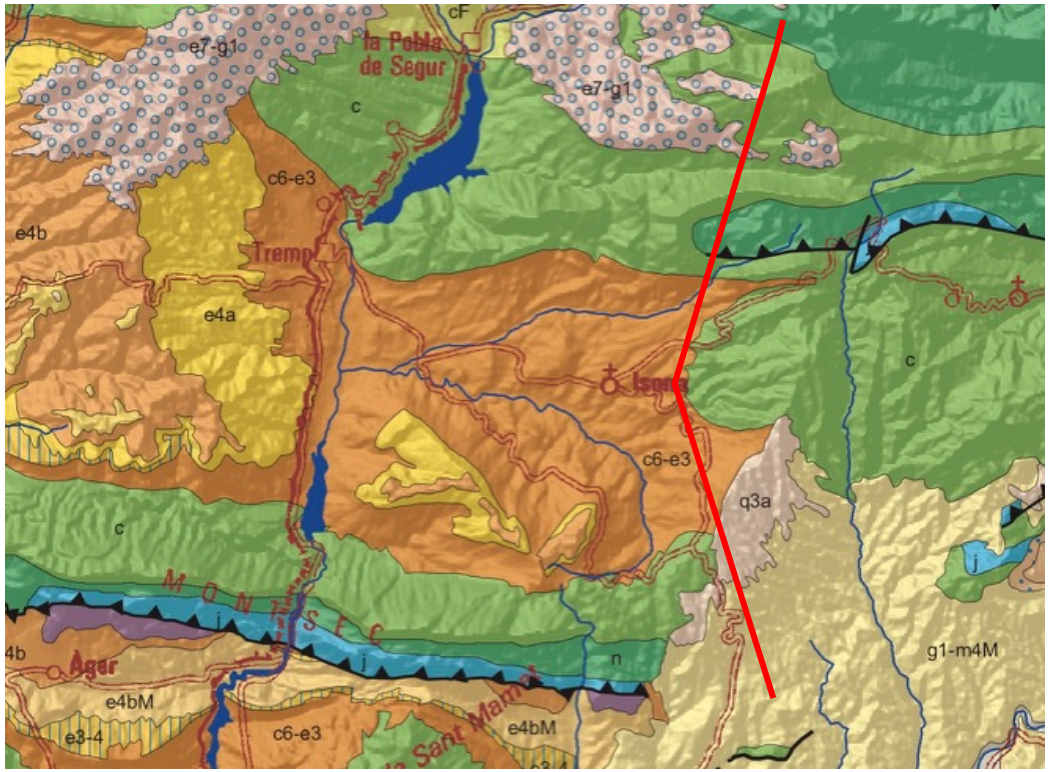
Overleaf shows a geological cross section through the south Pyrenean foreland basin in Spain (modified after Robert 2018), the geological column in a well drilled in the center, and the subsidence curve for that well. This data set allows an assessment of the petroleum system in the basin. The aim of the exercise is to complete the systems chart overleaf.

Tasks:
1. Using the stratigraphic column, identify the petroleum systems elements (source, reservoir, seal, overburden). There may be more than one of each.
2. Determine from the cross section the most likely traps, where they are, their type (stratigraphic vs structural), and when they formed – the latter information should go in the systems chart.
3. Draw the subsidence chart for the well, establish the thermal history of the well, and determine timings for oil/gas generation (see last week's exercise).
4. When was (were) the critical moment(s)?
5. Do you expect oil and/or gas in any reservoirs?

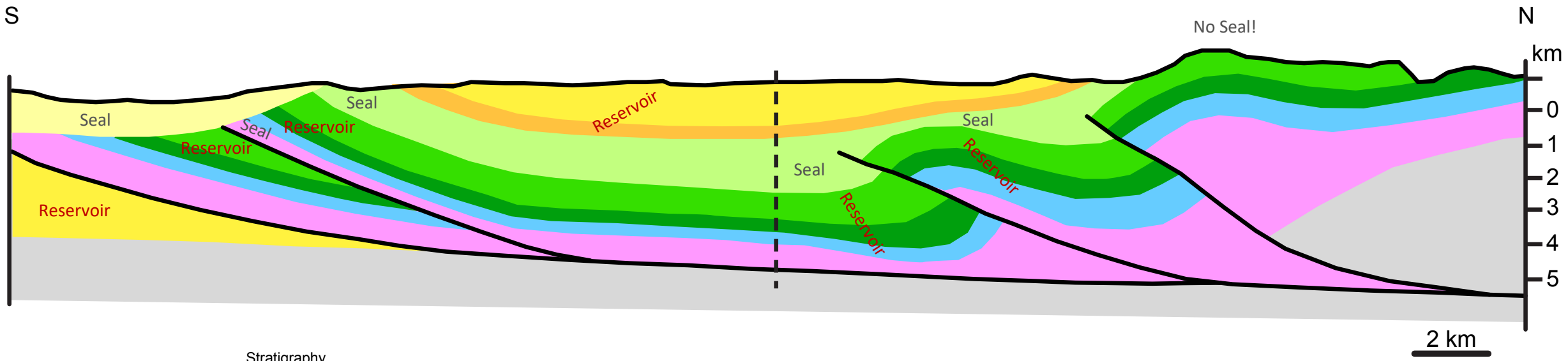


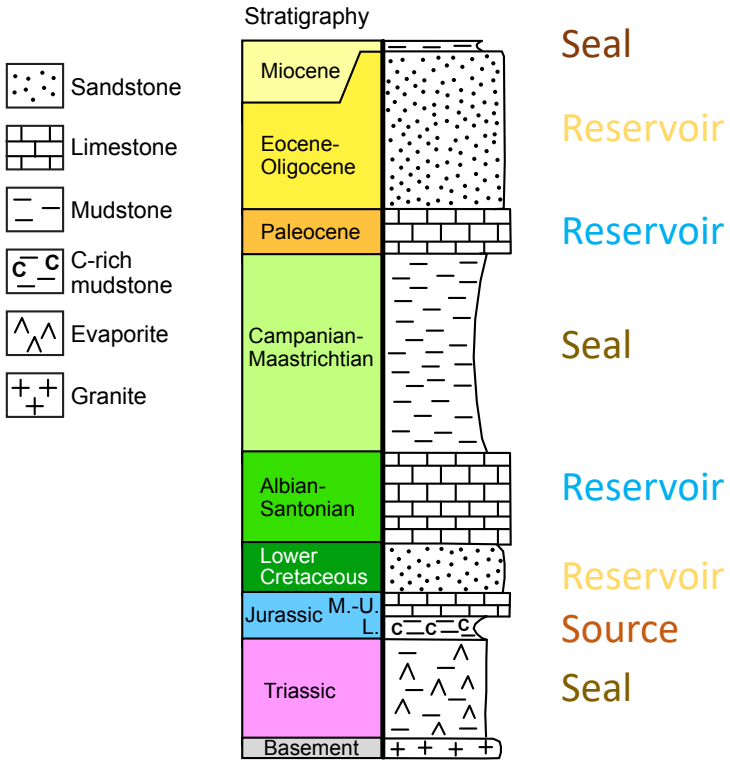


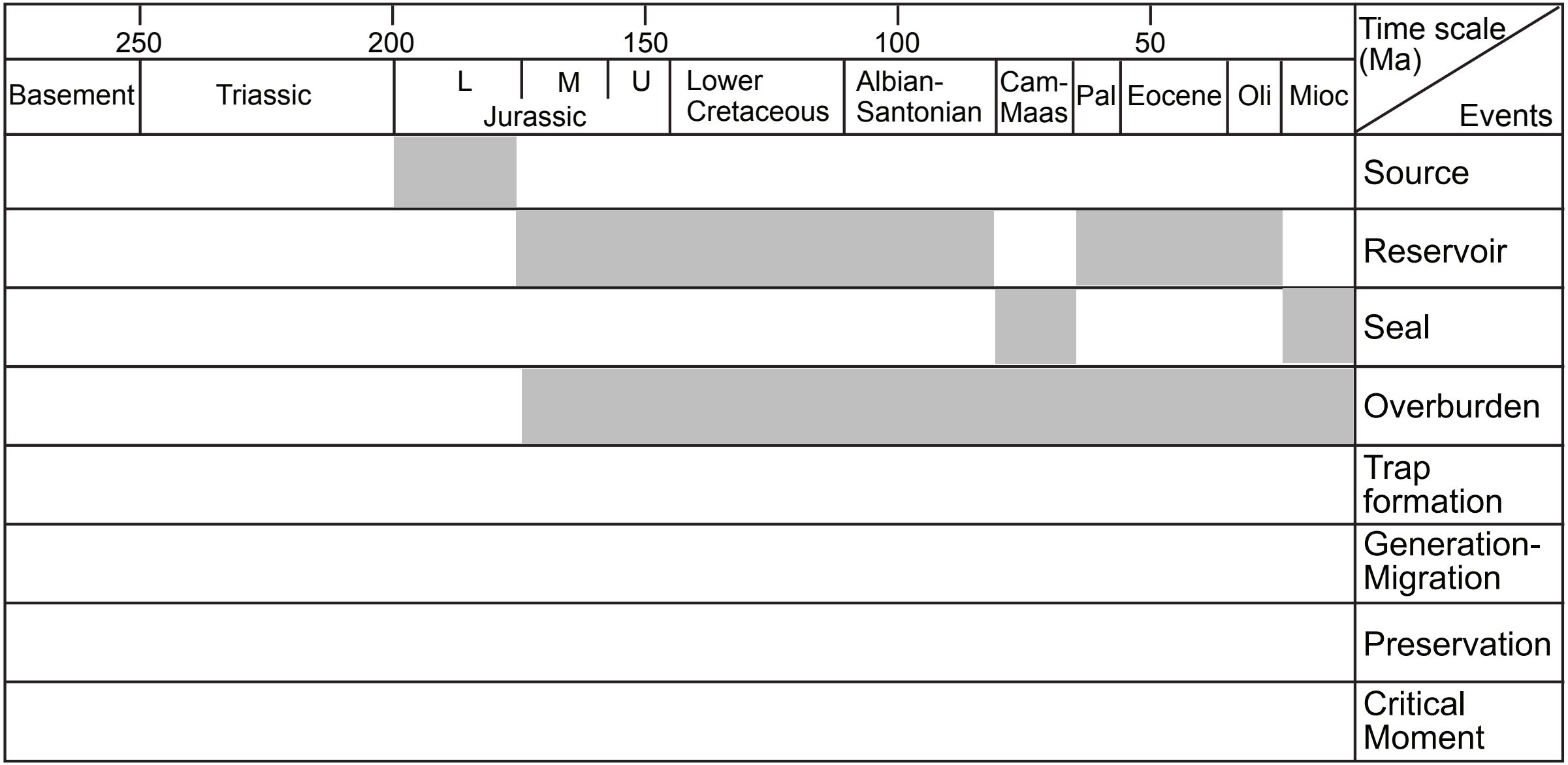
Geological map (colors not identical)

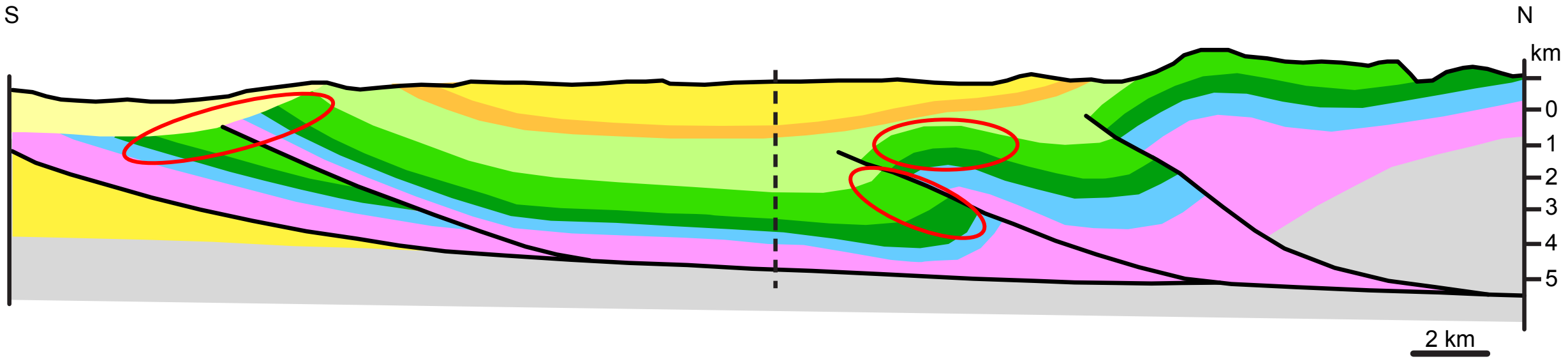


© IGC


S
N
No Seal!
km
0
1
2
3
4
5
Seal
Reservoir
Seal
Reservoir
Seal
Reservoir
Reservoir
Seal
Seal
Reservoir
Reservoir
2 km


Sandstone
Limestone
Mudstone
C-rich mudstone
Evaporite
Granite
Stratigraphy
Miocene
Eocene-Oligocene
Paleocene
Campanian-Maastrichtian
Albian-Santonian
Lower Cretaceous
Jurassic M.-U. L.
Triassic
Basement
Seal
Reservoir
Reservoir
Seal
Reservoir
Reservoir
Source
Seal

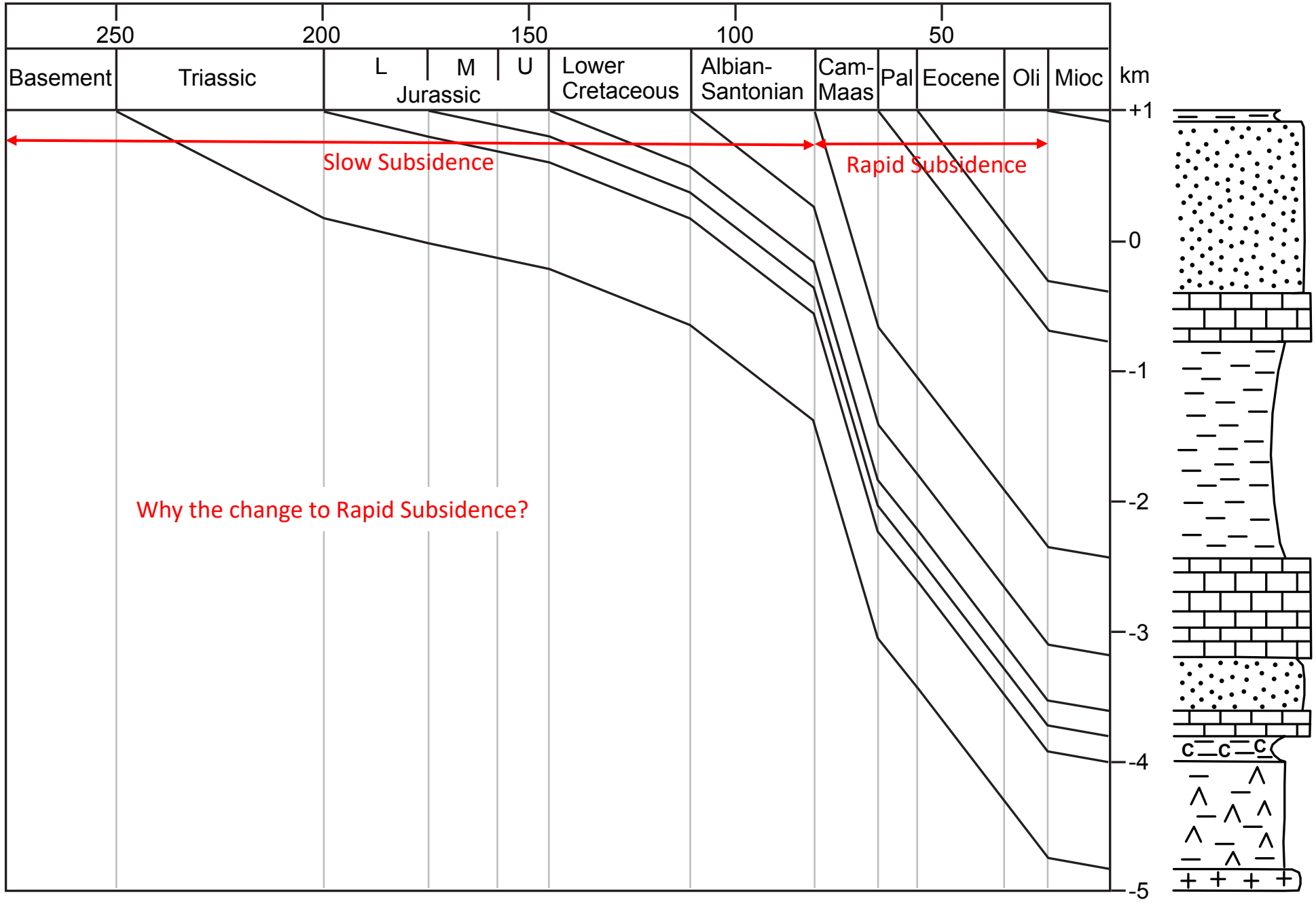

250
200
150
100
50
Basement
Triassic
L
M
U
Jurassic
Lower Cretaceous
Albian-Santonian
Cam-Maas
Pal
Eocene
Oli
Mioc
Time scale (Ma)
Events
Source
Reservoir
Seal
Overburden
Trap formation
Generation-Migration
Preservation
Critical Moment

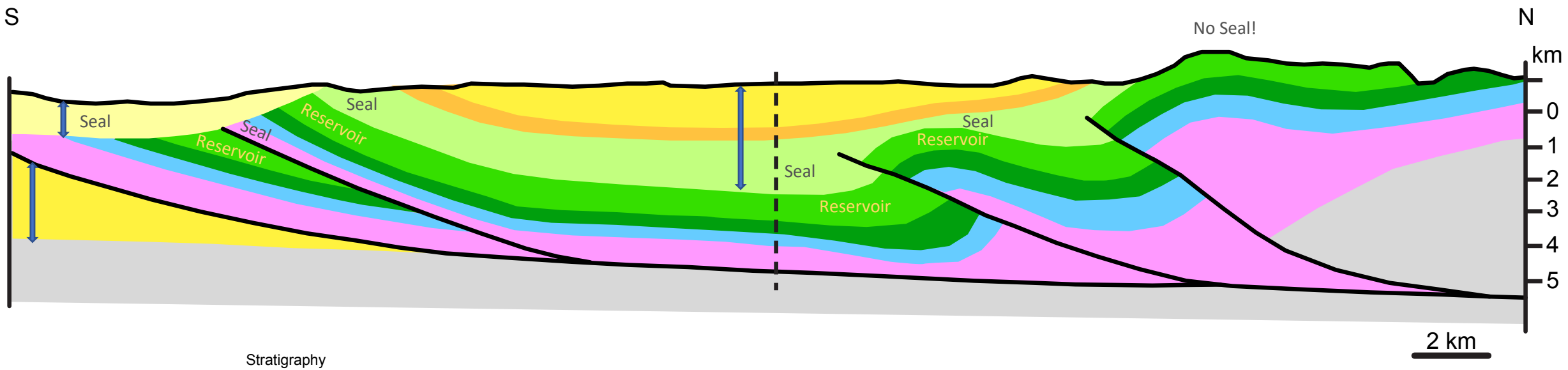


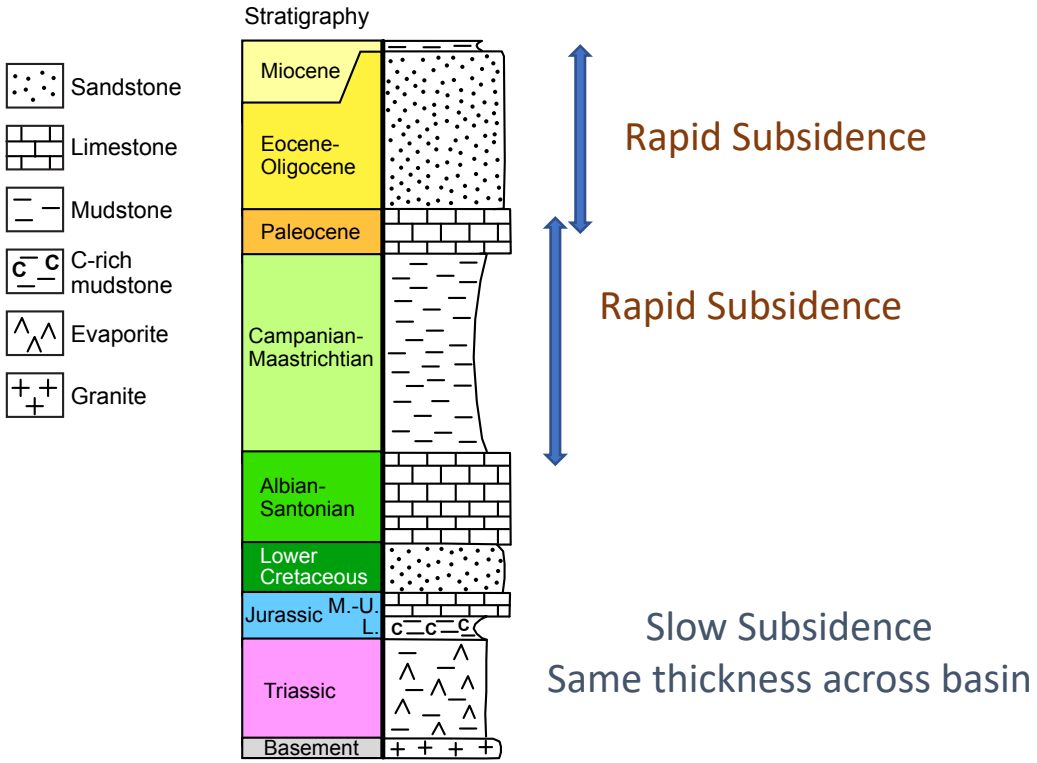
Stratigraphic (unconformity) trap
Formation Late Oligocene – Early Miocene

structural trap (requires sealing fault)
Formation Campanian

structural trap
Formation Campanian


250
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Basement
Triassic
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M
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Jurassic
Lower Cretaceous
Albian-Santonian
Cam-Maas
Pal
Eocene
Oli
Mioc
km
+1
0
-1
-2
-3
-4
-5
Slow Subsidence
Rapid Subsidence
Why the change to Rapid Subsidence?
C
C
C

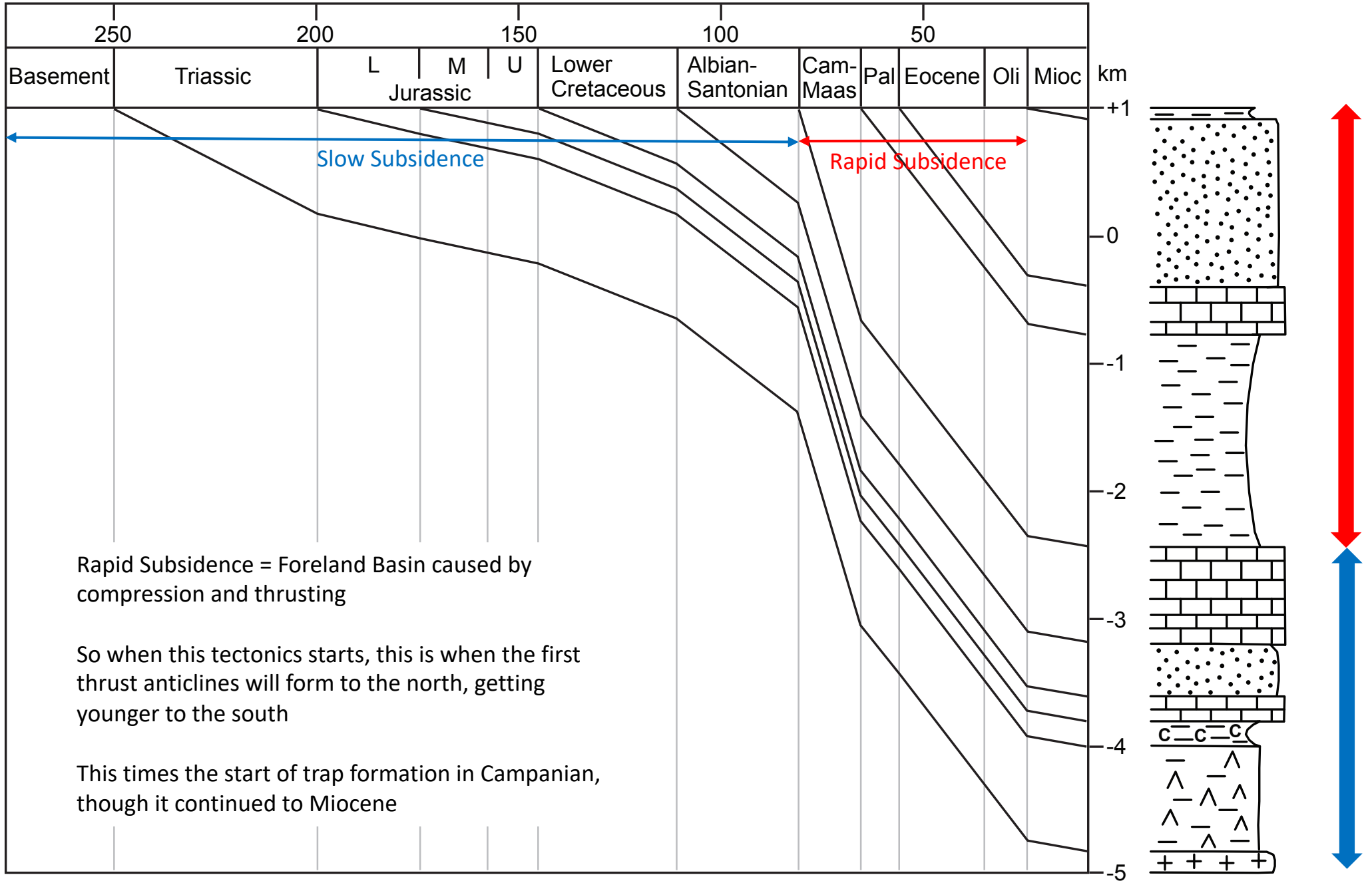


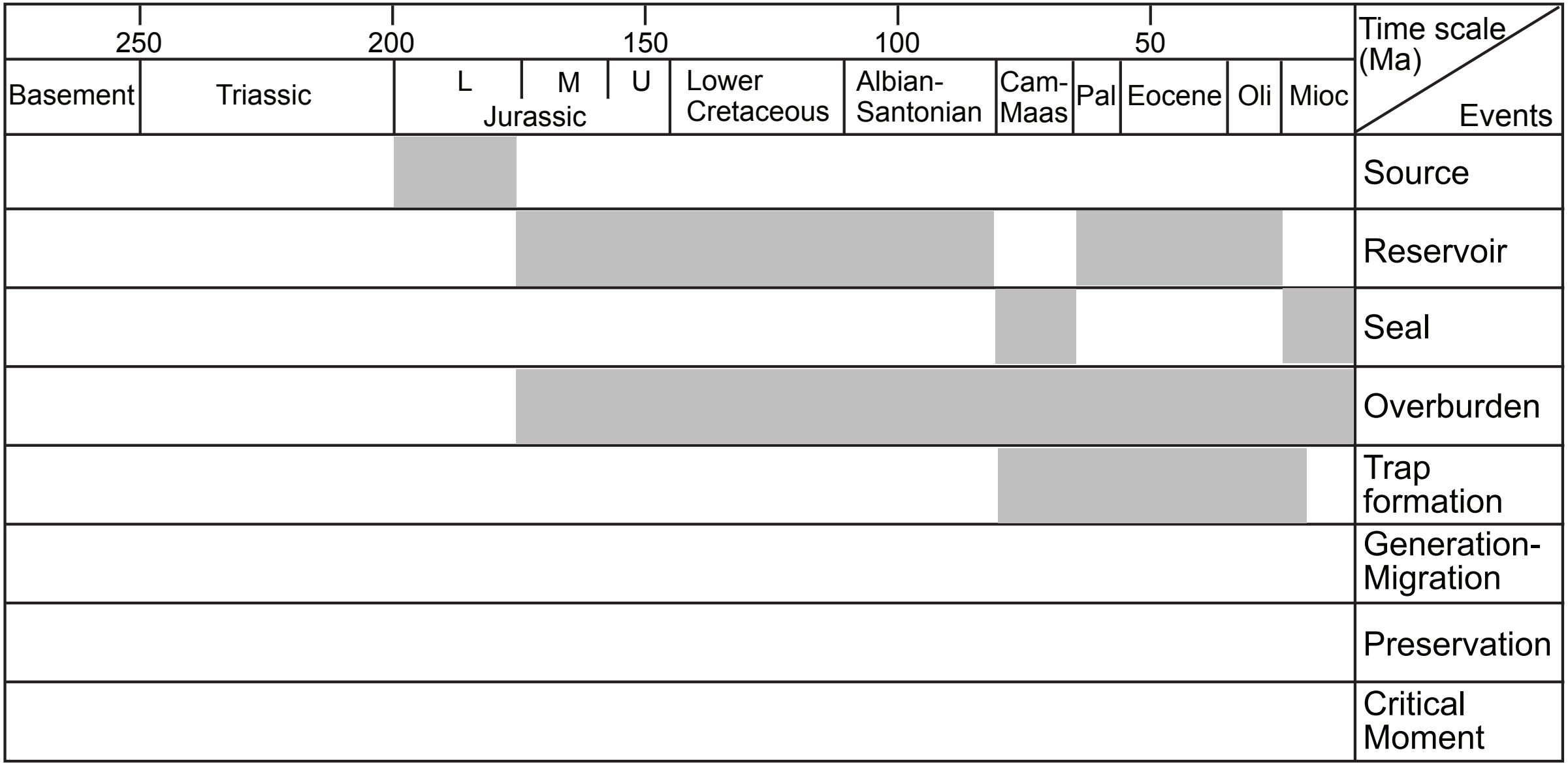


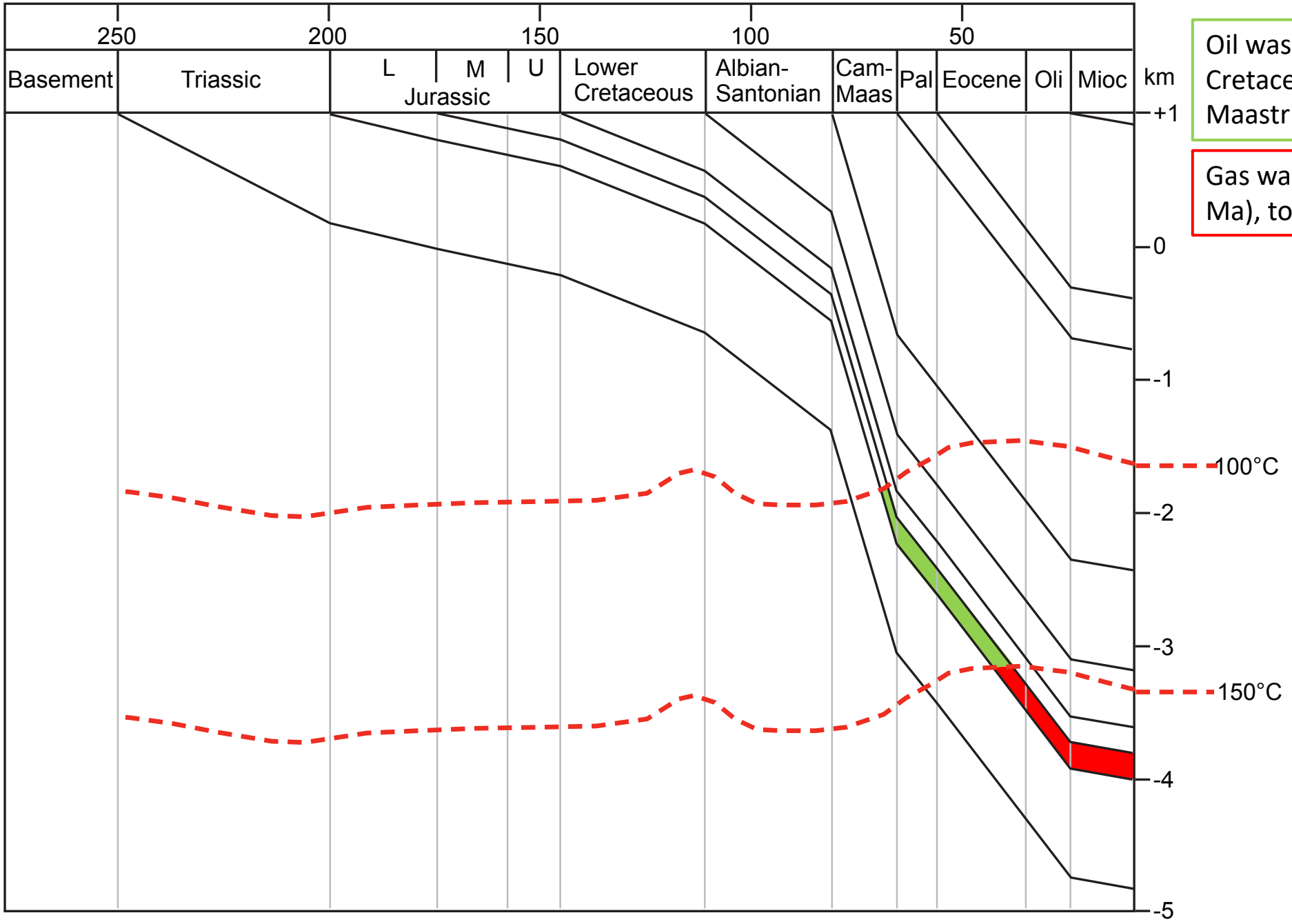
Rapid Subsidence

Rapid Subsidence

Slow Subsidence
Same thickness across basin

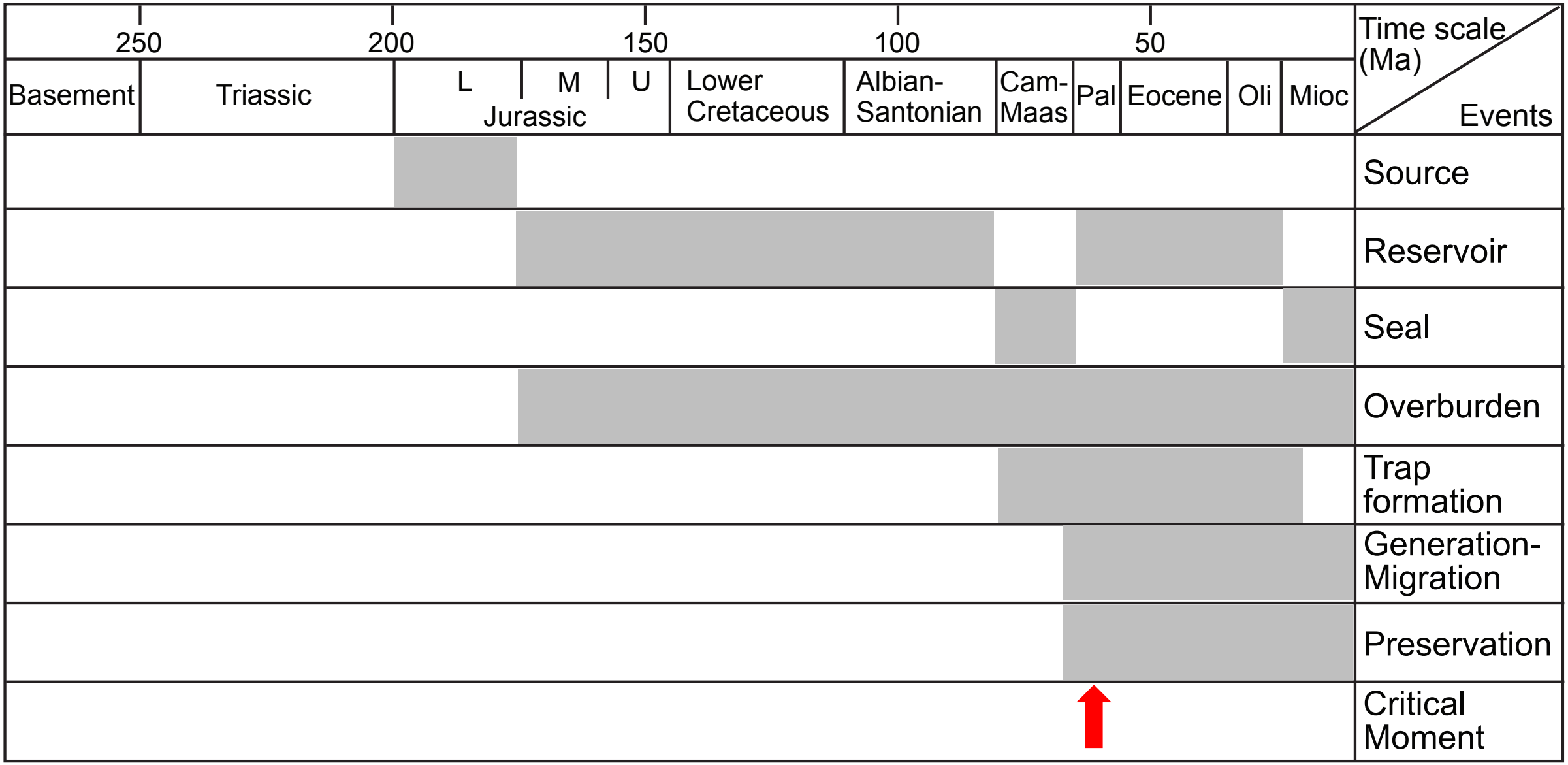

250
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Basement
Triassic
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Jurassic
Lower Cretaceous
Albian-Santonian
Cam-Maas
Pal
Eocene
Oli
Mioc
km
+1
0
-1
-2
-3
-4
-5
Slow Subsidence
Rapid Subsidence
Rapid Subsidence = Foreland Basin caused by compression and thrusting
So when this tectonics starts, this is when the first thrust anticlines will form to the north, getting younger to the south
This times the start of trap formation in Campanian, though it continued to Miocene

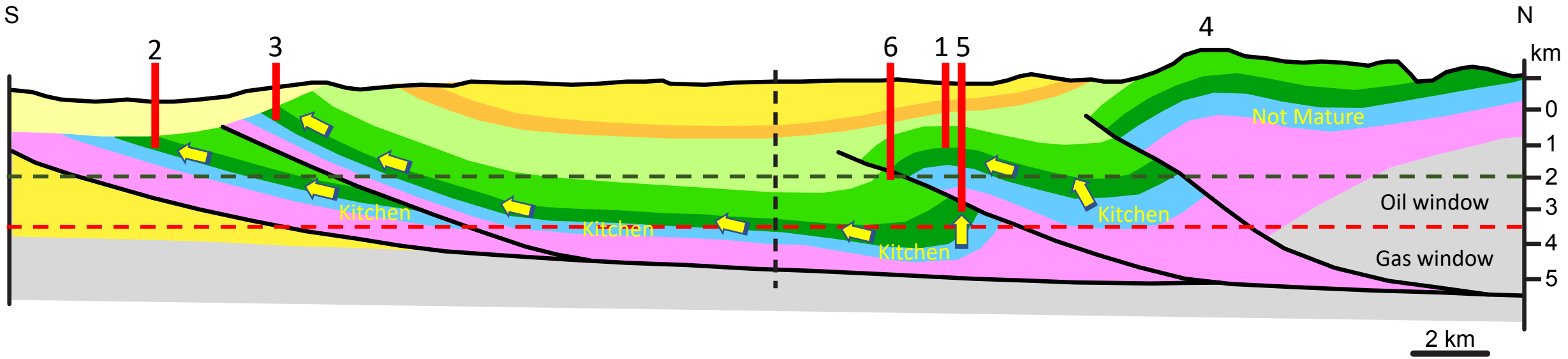

250
200
150
100
50
Basement
Triassic
L
M
U
Jurassic
Lower Cretaceous
Albian-Santonian
Cam-Maas
Pal
Eocene
Oli
Mioc
Time scale (Ma)
Events
Source
Reservoir
Seal
Overburden
Trap formation
Generation-Migration
Preservation
Critical Moment



Oil was formed since Late Cretaceous (~70 Ma, Maastrichtian), to Eocene (~40 Ma)

Gas was formed since Eocene (~40 Ma), to present day


250
200
150
100
50
Basement
Triassic
L
M
U
Jurassic
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Cam-Maas
Pal
Eocene
Oli
Mioc
Time scale (Ma)
Events
Source
Reservoir
Seal
Overburden
Trap formation
Generation-Migration
Preservation
Critical Moment



Traps / Plays

1. Thrust anticline, Albian-Santonian Reservoir, Camp/Maas Seal, Jurassic Source
2. Pinch out at Unconformity, Albian-Santonian Reservoir, Miocene Seal, Jurassic Source
3. Pinch out at Unconformity, Albian-Santonian Reservoir, Miocene Seal, Jurassic Source

4. Thrust anticline, Albian-Santonian Reservoir, Camp/Maas Seal missing, Jurassic Source not mature?
5. Sub Thrust trap, Lower Cretaceous Reservoir, Triassic Salt / Fault Seal, Jurassic Source – needs a Top Seal? in Albian?
6. Sub Thrust – no seal? Albian-Santonian Reservoir, Jurassic Source unless does fault seal?